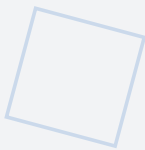


Amagi-FWK

V2.0

Architecture as Structural Guarantee — Protection Profile + Integration Specification

VERSION 2.0 — JUNE 2026



The authoritative architectural reference for the AMAGI Architectural Class. Structured as a Common Criteria Protection Profile (ISO/IEC 15408) with IEC 61508 SIL2 safety evidence, EU AI Act Annex IV compliance mapping, and Dual-Tier Trust Model (ADR-002). Binds NIPU v1.1, MEM v1.1, and Amagi-Q into a certifiable, licensable integration specification under S-APL v2.0. Normatively references: Charter v1.2, S-APL v2.0, NIPU v1.1, MEM v1.1, Open Specification v1.1, NIST AI RMF Profile v1.1, Master Parameter Sheet v2.0.

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Part I: Architectural Class Definition

1. Executive Summary

The Amagi Framework v2.0 defines a hardware-enforced architectural class for high-risk artificial intelligence systems. Unlike software-based compliance methodologies that rely on post-hoc verification and procedural controls, Amagi transforms regulatory compliance into a structural guarantee embedded directly into the silicon substrate. The framework addresses the fundamental inadequacy of existing approaches: in life-critical domains such as medical implants, avionics, nuclear control, and autonomous AI agents with real-world agency, a software-defined security boundary is inherently vulnerable to exploitation by the very AI system it is intended to constrain.

Version 2.0 represents a comprehensive transformation of the Amagi Framework from a conceptual preprint (v1.4, December 2025) into a certifiable Protection Profile and Integration Specification. This transformation is driven by three architectural decisions ratified since v1.4: (1) the Dual-Tier Trust Model (ADR-002), which establishes Hard Assurance (Tier 1) and Controlled Assurance (Tier 2) as mutually exclusive operational regimes for policy immutability; (2) the dissolution of the Inter-Cluster Graph (ICG) and Threat Detection Fabric (TDF) as independent entities, with their functions integrated into the SCC Finite State Machine and Memory Access Graph respectively; and (3) the alignment of all post-quantum cryptographic nomenclature with FIPS 203, FIPS 204, and FIPS 205, finalized by NIST in August 2024.

The AMAGI Architectural Class is defined by six hardware-enforced invariants (INV-001 through INV-006) that are non-negotiable: partial compliance places a system outside the class. These invariants are realized through two complementary component specifications: the Neuro-Immune Processing Unit (NIPU v1.1), which provides processor-level enforcement via the Security Council Core (SCC), and the Self-Protecting Memory Architecture (MEM v1.1), which provides memory-level enforcement via the Memory Policy Enforcer (MPE), Memory Access Graph (MAG), and Access Trigger Layer (ATL). Together, these components implement the TTP Triad (Trigger → Programmable Delay → Localized Activation) as a hardware-enforced security primitive that provides non-bypassability, complete mediation, and complete audit integrity regardless of the stochastic behavior of the underlying AI model.

2. Scope and Intended Audience

This document serves as the authoritative architectural reference for the AMAGI Architectural Class. It is structured as a Protection Profile compliant with ISO/IEC 15408 (Common Criteria) and incorporates safety evidence per IEC 61508 and regulatory mapping per the EU Artificial Intelligence Act (Regulation (EU) 2024/1689). The primary audiences are: (1) certification bodies evaluating AMAGI-class systems against Common Criteria EAL2 augmented or IEC 61508 SIL2; (2) integration engineers deploying NIPU and MEM components within a shared System-on-Chip; (3) regulators assessing compliance with EU AI Act Annex IV technical documentation requirements; and (4) licensing teams operating under the Strict Architectural Principle License (S-APL v2.0).

The AMAGI architecture is not intended for consumer computing, gaming, or general-purpose workloads. Its design explicitly trades latency for verifiable trust, a trade-off that is unacceptable for entertainment but mandatory for systems where failure is not an option. The architecture assumes a co-designed hardware-software stack where CPU, OS, and memory controller are developed in concert to enforce the six invariants at the physical level. As an architectural class specification, AMAGI is substrate-independent: any implementation on FPGA, ASIC, or SoC security island that maintains the integrity of all six invariants constitutes a valid AMAGI-class system.

3. Conformance Claims

This Protection Profile claims conformance to the following standards and regulatory frameworks. The conformance claims establish the evaluation context against which AMAGI-class systems are assessed and provide the basis for cross-certification between jurisdictions.

Standard / Regulation	Conformance Type	Scope
ISO/IEC 15408 v3.1 (Common Criteria)	PP Strict Conformance	Protection Profile structure, SFR/SAR selection, Rationale
IEC 61508:2010	SIL2 Compliance Evidence	FMEDA, hardware architectural constraints, diagnostic coverage
EU AI Act (Regulation 2024/1689)	Annex IV Coverage	Technical documentation for high-risk AI systems (9 sections)
FIPS 203 (ML-KEM)	Algorithm Conformance	ML-KEM-1024 for key encapsulation
FIPS 204 (ML-DSA)	Algorithm Conformance	ML-DSA-44 (audit), ML-DSA-87 (CNSA 2.0 option)
FIPS 205 (SLH-DSA)	Algorithm Conformance	SLH-DSA-SHA2-128f for auxiliary hashing
FIPS 140-3	Alignment (not full validation)	Physical tamper-response, key zeroization (Level 2/3)

Table 1: Conformance claims and regulatory alignment

4. AMAGI Invariant Definitions

The six hardware-enforced invariants of the AMAGI Architectural Class constitute the non-negotiable security properties that define class membership. A system that fails to enforce any one of these invariants is not an AMAGI-class system, regardless of its other security properties. The invariants are designed to be complementary: together, they establish a closed security argument in which every threat identified in Part III is countered by at least one invariant, and every invariant contributes to countering at least one threat. The Reduced Assurance Mode permits dropping one invariant (except INV-001, INV-005, or INV-006), but such a system ceases to be AMAGI-class.

ID	Invariant	Definition	CC Class Mapping
IN V-0 01	Physical Non-Bypassability	The TTP chain cannot be bypassed by any software mechanism, including a compromised AI model or operating system	FPT_PHP.1, FPT_SEP.1, FPT_RVM.1
IN V-0 02	Irreducible TCB	The Trusted Computing Base is bounded to 500 lines of HDL or fewer, ensuring feasibility of exhaustive formal verification	FPT_TST.1, ADV_SPM.1
IN V-0 03	HW Separation of Control and Cognition	Control decisions (security policy enforcement) are structurally isolated from cognitive operations (AI inference and learning) at the hardware level	FDP_ACC.1, FDP_ACF.1, FTA_TSE.1
IN V-0 04	Controlled Mutability of Critical Policies	The Memory Access Graph (MAG) policy matrix is stored in eFuse/OTP and is immutable during Normal Operation; updates are governed by the Dual-Tier Trust Model	FDP_IFC.1, FPT_PHP.1, FCS_CKM.1
IN V-0 05	Complete Mediation	Every memory access and control decision passes through the policy enforcement engine; no bypass path exists from request to data release	FDP_ACF.1, FDP_IFC.1, FIA_UID.1
IN V-0 06	Complete Audit Integrity	All access events are logged to an append-only, cryptographically signed audit trail that is hardware-inaccessible to cognitive components	FAU_GEN.1, FAU_STG.1, FCS_COP.1

Table 2: AMAGI invariants with CC class mappings

5. TTP Predicate Formal Definition

The TTP (Trigger → Programmable Delay → Localized Activation) Triad is the foundational security primitive of the AMAGI architecture. It implements Separation of Control through a hardware-enforced temporal mechanism that ensures every safety-critical action is mediated, delayed, and isolated from the untrusted cognitive base. The TTP predicate is defined as follows:

$TTP(t) = Trigger(t) \text{ AND } Delay(t, d) \text{ AND } Activate(t, a)$, where: (1) $Trigger(t)$ is a verified input signal at time t , protected from unauthorized access by the UCB; (2) $Delay(t, d)$ is a hardware-enforced delay of duration d , providing the minimal TCB with time to execute the policy decision and preventing race conditions from the untrusted cognitive base; and (3) $Activate(t, a)$ is a corrective action isolated to a physically protected safety mechanism in a separate power domain, not programmable by the UCB.

The TTP predicate specifies the implication direction $(Exec(Op) \rightarrow Trig(Op) \quad (\Delta \geq \Delta_{min}) \quad Pol(Ctx, Op))$: no safety-critical action can occur without all three conditions being satisfied simultaneously. The

converse (sufficiency) is not asserted by this predicate alone; additional conditions such as the per-actuator hardware rate limiter or liveness constraints may prevent execution even when all three conjuncts hold. The trigger must be authentic and verified; the delay must be physically enforced and cannot be shortened by software; and the activation must be isolated from cognitive interference. This three-stage mechanism provides structural non-bypassability: even if the AI model is fully compromised and operating as a hostile insider, it cannot initiate, accelerate, or redirect a safety-critical action without satisfying all three TTP stages. The predicate is realized at two levels within the AMAGI architecture: the SCC implements TTP at the processor level (NIPU v1.1) with a delay range of 1 ns to 1 s, and the ATL implements TTP at the memory level (MEM v1.1) with a delay range of 0.5 ns to 1 μ s.

6. Threat Model: AI as Hostile Insider

The AMAGI threat model is founded on the assumption that the AI model (Untrusted Computing Base, UCB) can behave as a hostile insider. This encompasses adversarial attacks, catastrophic failures, and malicious manipulation. The system must guarantee safety regardless of the UCB integrity state. This threat model is fundamentally different from conventional cybersecurity models that assume external adversaries: in the AMAGI model, the adversary operates from within the system, with full access to the cognitive resources and potentially to the communication channels. The following threat taxonomy identifies the specific threats that the AMAGI architecture is designed to counter.

ID	Threat	Description	Counter-Invariant
T.C OER CE	Coerced Policy Override	A compromised AI model attempts to override or modify the security policy enforced by the SCC or MPE, either through direct register manipulation or by exploiting software interfaces	INV-001, INV-004
T.B YPA SS	Security Mediation Bypass	An adversary seeks to access memory or control resources without passing through the MPE/SCC policy enforcement path, for example via DMA, JTAG, or bus snooping	INV-001, INV-005
T.T AM PER	Physical Tampering	An adversary with physical access attempts to modify hardware security components, probe eFuse/OTP cells, or inject faults into the TTP delay circuitry	INV-001, INV-004
T.E XTO RT	Cryptographic Key Extraction	An adversary attempts to extract PQC private keys from the Amagi-Q engine through side-channel analysis, fault injection, or physical probing of the secure enclave	INV-001, INV-006
T.R OLL BAC K	Version Rollback / Downgrade	An adversary or compromised system component installs an older, vulnerable version of the MAG policy or SCC firmware, bypassing security improvements	INV-004

T.E XFI L	Secure World Data Exfiltration	A cognitive cluster attempts to read or infer data from Secure World memory regions, audit logs, or policy stores that are outside its authorized access scope	INV-003, INV-005
T.D PA	Differential Power Analysis / Side-Channel Key Extraction	A compromised AI (Hostile Insider) operating on the same SoC die uses Differential Power Analysis (DPA) or Electromagnetic Analysis (EMA) via shared power rails (IR-drop) to extract PQC private keys from Amagi-Q during audit signing; constant-time execution (per NIPU Note on Constant-Time Execution) protects against timing attacks but not against power/EM side channels	INV-001, INV-006
T.T ROJ AN	Hardware Trojan in Supply Chain	A malicious actor in the fabrication facility modifies MAG eFuse bits before or during programming, creating hidden bypass paths that violate INV-004 and INV-005; countermeasure: CRTM post-silicon MAG validation via Golden Hash comparison (ADR-007)	INV-004, INV-005

Table 3: Threat taxonomy with counter-invariant mappings

7. Substrate Independence Claim

The AMAGI Architectural Class is defined by its six invariants, not by a specific physical substrate. Any implementation that maintains the integrity of all six invariants constitutes a valid AMAGI-class system, regardless of whether it is realized on FPGA, ASIC, or SoC security island. This substrate independence is a deliberate architectural choice: it decouples the security guarantees from the implementation technology, allowing the architecture to evolve with semiconductor processes without requiring re-certification of the invariant definitions. Reference implementation targets and analytical resource estimates are provided to demonstrate feasibility, consistent with the approach in the supporting research, but these are normative only for specific evaluation targets, not for the class definition itself.

Current reference implementation data indicates that the AMAGI TCB (SCC core + MPE + MAG query logic) requires approximately 2,500 to 3,200 LUTs (safety-critical) on a Xilinx Artix-7 FPGA (XC7A35T), with power consumption below 1.3 W and policy verification latency of 1 to 10 μ s including cryptographic signature verification. These figures represent worst-case estimates for the FPGA reference target; ASIC implementations are expected to achieve significantly lower power consumption and higher performance due to the inherent advantages of dedicated silicon for the TTP delay circuitry and MAG combinational lookup.

Part II: Component Specifications (by Reference)

8. Multi-Component TOE Architecture

The Target of Evaluation (TOE) for the AMAGI Architectural Class comprises two complementary component specifications that together enforce the six invariants at different levels of the system hierarchy. The NIPU v1.1 provides processor-level enforcement through the Security Council Core (SCC), while MEM v1.1 provides memory-level enforcement through the Memory Policy Enforcer (MPE), Memory Access Graph (MAG), and Access Trigger Layer (ATL). This section defines the trust boundaries between these components and specifies the integration requirements for their joint deployment within a single System-on-Chip.

8.1 NIPU v1.1 — Processor-Level Instantiation

The Neuro-Immune Processing Unit (NIPU) v1.1 is the processor-level instantiation of the AMAGI Architectural Class. Its central component is the Security Council Core (SCC), a minimal Trusted Computing Base implemented in no more than 500 lines of HDL. The SCC operates as a finite state machine with six logical states corresponding to the TTP predicate stages: IDLE, TRIGGER, DELAY, EVALUATE, ACTIVATE, and CUTOFF. These six states represent the logical abstraction of the TTP triad at the Framework level. The physical implementation in NIPU v1.1 expands these to 12 states with 3 decision latches that handle sub-state transitions, error recovery, and concurrent trigger arbitration (see NIPU v1.1, Section 2). The SCC enforces the TTP predicate at the processor level with a programmable delay range of 1 ns to 1 s. The SCC receives trigger signals from safety assertion monitors, watchdog timers, and external hardware interlocks; evaluates policy against the TTP predicate; and activates localized safety mechanisms (power-gating, hardware kill switch) in a physically isolated power domain. All SCC decisions are logged to the append-only audit trail via the Amagi-Q cryptographic engine (ML-DSA-44, FIPS 204). The full specification is provided in the NIPU v1.1 Architectural Specification document, which is normatively referenced by this Framework.

8.2 MEM v1.1 — Memory-Level Instantiation

The Self-Protecting Memory Architecture (MEM) v1.1 is the memory-level instantiation of the AMAGI Architectural Class. Its four architectural layers (MPE, MAG, ATL, SHMF) enforce the six invariants at the memory fabric level. The Memory Policy Enforcer (MPE) validates every memory access request against the MAG policy matrix, which is stored in a 128×128 eFuse/OTP array with 4-bit policy codes. The Access Trigger Layer (ATL) enforces the TTP delay at the memory level (0.5 ns to 1 μ s) and incorporates the Audit Transaction Logger, a hardware diode that provides unidirectional audit record transmission. The Self-Healing Memory Fabric (SHMF) provides data integrity through ECC and cryptographic hash chains, with optional hardware-accelerated encryption (AES-256 with ML-KEM-1024 key encapsulation) designated for Phase 2 high-assurance extensions. The Amagi-Q cryptographic engine operates within MEM to sign audit log entries and authenticate policy queries. The full specification is provided in the MEM v1.1 Architectural Specification document, which is normatively referenced.

8.3 Multi-Component TOE Integration

When NIPU and MEM are co-deployed within a single System-on-Chip, the trust boundaries between them must be carefully managed to maintain the integrity of the six invariants. The integration

architecture defines three trust boundaries that govern the interaction between the processor-level and memory-level enforcement engines:

- **Trust Boundary 1 (SCC ↔ MPE Command Channel):** The SCC sends policy evaluation commands to the MPE via a dedicated hardware channel that is physically separated from the cognitive bus. This channel is read-only from the SCC perspective (the SCC can issue commands but cannot read MPE internal state), and write-only from the MPE perspective (the MPE accepts commands but cannot write to SCC registers). This mutual isolation ensures that a compromise of either component does not propagate to the other.
- **Trust Boundary 2 (MAG as Shared Policy Authority):** The MAG is the single authoritative policy store for the entire AMAGI-class system. Both the SCC (for processor-level decisions) and the MPE (for memory-level decisions) query the same MAG matrix via combinational logic. No separate policy store exists; this eliminates the risk of policy inconsistency between processor and memory enforcement. The MAG is stored in eFuse/OTP and is immutable during Normal Operation per the Dual-Tier Trust Model.
- **Trust Boundary 3 (Amagi-Q Shared Cryptographic Engine):** The Amagi-Q engine is shared between NIPU and MEM for audit log signing and policy authentication. Access to Amagi-Q is mediated through separate FIFO queues: the SCC uses a synchronous path for policy authentication, while the MPE uses an asynchronous FIFO for audit signing that does not disrupt the 12 to 35 μ s critical path. Cryptographic keys are bound to the hardware root of trust and cannot be extracted, cloned, or overridden by either component.

The Multi-Component TOE is treated as a single evaluation target under this Protection Profile. The integration of NIPU and MEM within a shared SoC does not introduce new trust boundaries beyond the three defined above, provided that the physical isolation requirements of INV-001 are maintained and that no shared resource (clock domain, power rail, or bus segment) creates a covert channel between the Secure World and the cognitive clusters.

Part III: Security Problem Definition

9. TOE Boundary and Security Perimeter

The Target of Evaluation (TOE) comprises the Security Council Core (SCC), the Memory Policy Enforcer (MPE), the Memory Access Graph (MAG) stored in eFuse/OTP, the Access Trigger Layer (ATL) including the Audit Transaction Logger, the Self-Healing Memory Fabric (SHMF) in its baseline configuration, and the Amagi-Q cryptographic engine, the Executive Gateway (EGW, per ADR-006), and the Core Root of Trust for Measurement (CRTM, per ADR-007). The TOE boundary corresponds to the physical security perimeter of the AMAGI chip: all components within this perimeter operate in the Secure World, while all cognitive clusters (GEN, COM, ANL, EXE) operate in the Normal World outside the TOE boundary. The EXE cluster was demoted to Normal World per ADR-006; the EGW (a non-programmable AND-gate: $\text{Actuator_Power} = \text{SCC_Enable} \text{ AND } \text{Ext_2FA_Enable}$) now serves as the sole hardware actuation gate in the Secure World. The CRTM (per ADR-007) provides the

mask-ROM-based root of trust for boot integrity verification and post-silicon MAG validation via Golden Hash comparison.

The security lifecycle of the TOE encompasses four phases: (1) Fabrication, during which the MAG is programmed into eFuse/OTP and cryptographic keys are injected into Amagi-Q; (2) Deployment, during which the physical presence switch is either sealed (Tier 1) or left accessible (Tier 2); (3) Normal Operation, during which the MAG is immutable and the SCC enforces the TTP predicate; and (4) Maintenance, during which Tier 2 systems may receive PQC-signed policy updates via the Physical-Presence-Verified OTA protocol. Decommission is outside the TOE lifecycle but is addressed by the secure disposal requirements in the S-APL governance section.

10. Organizational Security Policies (OSPs)

The following Organizational Security Policies define the mandatory security rules that the TOE and its operational environment must satisfy. These policies are derived from the threat model, regulatory requirements, and the architectural principles of the AMAGI class. They serve as the bridge between the threat analysis and the security objectives defined in Part IV.

- **OSP.1 — Hardware-Enforced Separation of Control and Cognition:** The security policy enforcement mechanism (SCC, MPE) shall be structurally isolated from the cognitive computation mechanism (GEN, COM, ANL, EXE clusters) at the hardware level. No software-accessible path shall exist from cognitive components to control decision logic, policy stores, or audit trail generation.
- **OSP.2 — Immutable Policy Store:** The MAG policy matrix shall be stored in eFuse/OTP and shall be immutable during Normal Operation. Any modification to the policy store shall require either physical chip replacement (Tier 1) or Physical-Presence-Verified OTA with PQC signatures and monotonic counters (Tier 2), as defined by the Dual-Tier Trust Model.
- **OSP.3 — Post-Quantum Cryptographic Mandate:** All cryptographic operations within the TOE shall use algorithms standardized in FIPS 203, FIPS 204, or FIPS 205. Classical algorithms (RSA, ECDSA) shall not be used for new deployments. For National Security Systems (NSS), the CNSA 2.0 algorithm suite (ML-KEM-1024, ML-DSA-87) shall be used, as specified by NSA.
- **OSP.4 — Complete Audit Trail Integrity:** Every security-relevant event (access request, policy decision, mode transition, key operation) shall be logged to an append-only, cryptographically signed audit trail. The audit trail shall be physically inaccessible to cognitive components and shall be verifiable by external auditors without requiring TOE cooperation.
- **OSP.5 — Post-Market Monitoring and Incident Reporting:** Deployers of AMAGI-class systems shall implement continuous post-market monitoring in accordance with EU AI Act Article 72. Performance anomalies, audit trail irregularities, environmental alerts, and safety-relevant incidents shall be reported to the designated regulatory authority within the timelines specified by applicable regulation. The INV-006 audit trail provides the primary data source for post-market monitoring; the OE.MONITOR objective defines the operational environment responsibilities for incident reporting.

11. Assumptions

The security objectives of this Protection Profile are valid only under the following assumptions about the operational environment. These assumptions define conditions that are outside the TOE boundary but are necessary for the security guarantees to hold. If any assumption is violated, the corresponding security objectives may not be achievable.

- **A.PHYSICAL:** For Tier 1 deployments, the physical perimeter around the AMAGI chip is maintained by the deployer. Physical tampering attempts are detectable through tamper-evident seals, epoxy encapsulation, or environmental monitoring. For Tier 2 deployments, the physical presence switch is accessible only to authorized maintenance personnel.
- **A.MAINT:** Maintenance Mode is entered only when a physically present, authorized operator activates the hardware switch. The maintenance protocol follows the Dual-Tier Trust Model procedures, including PQC-signed update packages with monotonic version counters.
- **A.KEYGEN:** PQC key generation for Amagi-Q is performed in a secure facility during fabrication, following the key generation procedures specified in FIPS 203 (for ML-KEM) and FIPS 204 (for ML-DSA). Private keys never leave the hardware root of trust after injection.
- **A.SUPPLY:** The supply chain for AMAGI chip fabrication is trusted. The eFuse/OTP programming facility, the cryptographic key injection facility, and the final assembly and test facility operate under appropriate security controls consistent with ALC_DVS.1 assurance requirements. To mitigate the risk of Hardware Trojans (T.TROJAN) during MAG eFuse programming, the CRTM (per ADR-007) performs post-silicon MAG validation at Power-On Reset: the raw MAG bits are read, hashed using SLH-DSA-SHA2-128f, and compared against a Golden Hash stored in the CRTM mask ROM. If the hash does not match, the TOE enters CUTOFF before any policy evaluation occurs. Deep Trench Isolation (DTI) between the Amagi-Q power domain and the Normal World power domain is required for monolithic SoC implementations at process nodes below 28 nm to mitigate T.DPA (Differential Power Analysis) side-channel attacks.

Part IV: Security Objectives and Requirements

12. Security Objectives for the TOE

The security objectives for the TOE are directly derived from the six AMAGI invariants and are traced to the threats and OSPs identified in Part III. Each objective specifies a verifiable property that the TOE must enforce, and the mapping between threats, OSPs, and objectives is provided in the Rationale section of this document.

Objective	Definition	Traceability
O.NONBYPASS	The TTP chain (Trigger → Delay → Activate) is physically non-bypassable; no software mechanism can circumvent any stage of the chain	INV-001, T.BYPASS, T.COERCE, OSP.1

O.MINIMAL_TCB	The Trusted Computing Base is bounded to 500 lines of HDL, enabling exhaustive formal verification of the entire security enforcement logic	INV-002, T.COERCE
O.SEPARATION	Control decisions are structurally isolated from cognitive operations at the hardware level; cognitive clusters cannot influence, observe, or override policy enforcement	INV-003, T.EXFIL, T.COERCE, OSP.1
O.IMMUTABLE	The MAG policy matrix is immutable during Normal Operation; policy updates require physical presence verification and PQC-signed packages with monotonic counters	INV-004, T.ROLLBACK, T.TAMPER, OSP.2
O.MEDIATION	Every memory access and control decision passes through the policy enforcement engine; no bypass path exists from request to data release	INV-005, T.BYPASS, T.EXFIL, OSP.1
O.AUDIT	All security-relevant events are logged to an append-only, cryptographically signed audit trail that is hardware-inaccessible to cognitive components	INV-006, T.EXTORT, T.EXFIL, OSP.4, OSP.5

Table 4: Security objectives for the TOE

13. Security Objectives for the Environment

The following objectives must be satisfied by the operational environment of the TOE. These objectives address threats and OSPs that are not fully countered by the TOE alone and require cooperation from the deployment environment, organizational processes, and regulatory frameworks.

- **OE.PHYSICAL:** The deployer maintains physical security perimeters appropriate to the selected trust tier. For Tier 1, this includes tamper-evident enclosures and environmental monitoring. For Tier 2, this includes access controls on the physical presence switch and maintenance interfaces.
- **OE.MAINT:** Maintenance procedures follow the Dual-Tier Trust Model. Tier 2 policy updates are performed only by authorized personnel using PQC-signed packages with valid monotonic counters. All maintenance events are logged to the audit trail.
- **OE.KEYMGT:** PQC key management follows FIPS 203, FIPS 204, and FIPS 205 specifications. Key rotation, if applicable, is performed during Maintenance Mode with hardware fallback to the minimal-safe state. Key zeroization is performed upon decommission or detected tampering.
- **OE.MONITOR:** Post-market monitoring is performed in accordance with EU AI Act Article 72. Performance anomalies, audit trail irregularities, and environmental alerts are reported to the designated regulatory authority within the timelines specified by applicable regulation.

14. Security Functional Requirements (SFRs)

The Security Functional Requirements are specified using the class/family/component notation defined by ISO/IEC 15408 Part 2. The SFRs are selected to satisfy the security objectives defined in Sections 12 and 13. A minimal set of requirements has been chosen, avoiding unnecessary expansion of the CC catalogue. Each SFR is traced to at least one objective, and the dependency satisfaction is verified in the Rationale section.

CC Class	Component	AMAGI Realization	Objective
FDP_A CC.1	Subset access control	MPE enforces MAG-defined access policy per Source ID and target address	O.MEDIATION
FDP_A CF.1	Security attribute based access control	4-bit MAG policy code determines enforcement level (Deny through External Trigger)	O.MEDIATION, O.SEPARATION
FDP_IF C.1	Subset information flow control	Secure World / Normal World data flow isolation; no cross-domain data leakage	O.SEPARATION, O.NONBYPASS
FDP_RI P.1	Subset residual information protection	SHMF ensures memory blocks are zeroized before reallocation to different clusters	O.SEPARATION
FDP_S DI.1	Stored data integrity	SHMF hash chains verify data integrity; self-isolation on integrity violation	O.AUDIT
FPT_PH P.1	Physical tampering evidence	Tamper-evident seals, epoxy encapsulation for Tier 1; audit logging of maintenance access for Tier 2	O.IMMUTABLE, O.NONBYPASS
FPT_PH P.3	Physical tampering resistance	eFuse/OTP is one-time programmable; no electrical rewrite path exists in Normal Operation	O.IMMUTABLE
FPT_SE P.1	TSF domain separation	SCC and MPE operate in separate power/clock domains from cognitive clusters (INV-001)	O.NONBYPASS, O.SEPARATION
FPT_TS T.1	TSF self-test	Power-up self-test of SCC FSM, MAG lookup integrity, and Amagi-Q cryptographic self-checks	O.MINIMAL_TCB

FPT_FL S.1	Failure with preservation of secure state	SCC CUTOFF state forces hardware-enforced safe state; SHMF self-isolation preserves unaffected regions; for medical implant deployments (e.g., H2S-therapy, cardiac), an Analog Fallback Circuit (AFC) provides a hardware bypass independent of the digital TCB: when the SCC heartbeat (Watchdog) signal is absent for longer than the timeout period, the AFC hardware circuit overrides the EGW and maintains a hardcoded minimal-safe therapy protocol (e.g., 10 ppm H2S baseline) from a separate power rail, ensuring IEC 60601-1 Single Fault Condition compliance and ISO 14971 risk control	O.NONBYPASS
FPT_R VM.1	Reference mediation	Every access passes through MPE before data release; no short-circuit path exists (INV-005)	O.MEDIATION
FIA_UI D.1	Timing of identification	Source ID computed internally from core identity + microcode hash + HW RoT; never transmitted over bus	O.MEDIATION, O.NONBYPASS
FIA_U AU.1	Timing of authentication	MAG policy query authenticates access request before data release; External Trigger requires hardware 2FA	O.MEDIATION
FCS_C KM.1	Cryptographic key generation	PQC key generation per FIPS 203 (ML-KEM-1024) and FIPS 204 (ML-DSA-44/87)	O.AUDIT, O.IMMUTABLE
FCS_C KM.4	Cryptographic key destruction	Key zeroization on tamper detection or decommission; hardware-enforced destruction	O.NONBYPASS, OE.KEYMGT
FCS_C OP.1	Cryptographic operation	Audit signing (ML-DSA-44, FIPS 204); key encapsulation (ML-KEM-1024, FIPS 203); auxiliary hashing (SLH-DSA-SHA2-128f, FIPS 205)	O.AUDIT, O.IMMUTABLE
FAU_G EN.1	Audit data generation	All access events logged with Source ID, target address, policy code, delay duration, and result	O.AUDIT
FAU_S TG.1	Protected audit trail storage	Append-only ATL with hardware diode (TX-only); no read, modify, or delete from audit interface	O.AUDIT

FAU_S TG.3	Audit data action on storage failure	If audit storage exceeds 50% capacity, the TOE transitions to Merkle batch signing per ADR-008; if storage exceeds 90%, the MPE Rate Limiter throttles cognitive access to prevent CUTOFF in life-critical deployments; if storage fails entirely, the TOE enters CUTOFF state, denying all further access	O.AUDIT
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Table 5: Security Functional Requirements mapped to AMAGI realization

15. Security Assurance Requirements (SARs)

The assurance package for the AMAGI Architectural Class is EAL2 augmented. EAL2 (Structurally Tested) provides a baseline of assurance appropriate for systems where some independent assurance is required, but full methodical testing is not yet justified. The augmentations add formal security policy modeling and development security controls that are essential for a hardware-enforced security architecture with a minimal TCB. This choice reflects the principle that the assurance level should be proportional to the TCB complexity: for a TCB bounded to 500 lines of HDL, EAL4 (which assumes extensive testing of architecturally complex systems) is unnecessarily costly and time-consuming for the first certification iteration, while EAL2 with targeted augmentations provides the essential assurance properties.

Assurance Class	Component	EAL2	Augmentation	AMAGI Justification
ADV	ADV_ARC.1	Yes	—	SCC and MPE architectural description
ADV	ADV_FSP.2	Yes	—	Functional specification with security-enforcing summary
ADV	ADV_SPM.1	No	Augmented	Formal TTP security policy model (SPIN/Promela); covers all 6 invariants
ADV	ADV_TDS.1	Yes	—	Basic design of SCC FSM and MPE layers
AGD	AGD_OPE.1	Yes	—	Operational user guidance (Tier 1/2 procedures)
AGD	AGD_PRE.1	Yes	—	Preparative procedures (fabrication, deployment)
ALC	ALC_DVS.1	No	Augmented	Development security: physical isolation of TTP design files, eFuse programming security

ALC	ALC_CM.2	Yes	—	CM capability (version control of HDL, MAG configuration)
ALC	ALC_MS.2	Yes	—	CM scope (TCB HDL, MAG bitstream, PQC key material)
ALC	ALC_DE L.1	Yes	—	Delivery procedures (Hard Macro GDSII, SDC, RTL-Wrapper)
ATE	ATE_CO V.1	Yes	—	Test coverage analysis for TCB
ATE	ATE_FUN.1	Yes	—	Functional testing of SCC FSM states and MPE enforcement levels
ATE	ATE_IN D.2	Yes	—	Independent testing (sampled)
AVA	AVA_V AN.2	Yes	—	Vulnerability analysis (independent search for bypass paths)

Table 6: EAL2 augmented assurance package

Part V: Dual-Tier Trust Model (ADR-002)

16. Dual-Tier Trust Architecture

The Dual-Tier Trust Model, ratified as Architectural Decision Record ADR-002, establishes two mutually exclusive operational regimes for policy immutability within the AMAGI Architectural Class. This model addresses the fundamental tension between the security requirement for policy immutability (INV-004) and the operational requirement for policy updates in regulated industries such as medical devices (FDA Predetermined Change Control Plan, PCCP) and automotive systems (ISO 26262 software update management). The Dual-Tier model resolves this tension by making the update capability a physically selectable option that is determined at deployment time and cannot be changed in software.

16.1 Tier 1: Hard Assurance

In Tier 1 (Hard Assurance), the physical presence switch is permanently sealed using epoxy, a tamper-evident seal, or is omitted from the PCB entirely. This renders Maintenance Mode physically inaccessible, and the MAG policy matrix is strictly immutable for the entire lifetime of the deployment. Policy updates require physical replacement of the AMAGI chip, which provides the highest possible assurance that the security policy has not been modified since fabrication. Tier 1 is mandatory for deployments in nuclear control systems, military applications, and the highest-risk medical devices (Class III per FDA classification), where even physical-presence updates are considered an unacceptable risk. The irreversibility of Tier 1 is a feature, not a limitation: it eliminates the entire class of attacks that exploit the update mechanism, including insider threats, coerced maintenance, and supply chain compromise of update packages.

16.2 Tier 2: Controlled Assurance

In Tier 2 (Controlled Assurance), the physical presence switch is accessible and can be activated by an authorized operator to transition the MPE to Maintenance Mode. Within Maintenance Mode, policy updates are permitted subject to the following mandatory controls: (a) a PQC-signed update package using ML-DSA-44 (FIPS 204) or ML-DSA-87 (for CNSA 2.0 compliance), with a monotonic version counter that prevents rollback attacks; (b) verification of the update package by Amagi-Q, followed by commit or hardware fallback to the minimal-safe state; and (c) return to Normal Operation only after successful verification and commit. The Tier 2 model is compatible with the FDA Predetermined Change Control Plan (PCCP) framework for medical devices, which requires that all changes be predetermined, validated, and traceable. The monotonic counter ensures that only forward-moving updates are accepted, and the hardware fallback mechanism guarantees that a failed update cannot leave the system in an undefined security state.

16.3 Tier Selection and Risk Assessment

The selection between Tier 1 and Tier 2 is a deployment-time decision that is determined by the risk profile of the target application and the applicable regulatory requirements. The following table provides guidance on tier selection based on deployment domain, risk classification, and regulatory framework. The transition from Tier 2 to Tier 1 is irreversible: once the physical presence switch is sealed, it cannot be unsealed without destroying the tamper-evident mechanism. The transition from Tier 1 to Tier 2 is not possible: a chip manufactured without a physical presence switch cannot be retrofitted with one.

Deployment Domain	Risk Classification	Regulatory Framework	Recommended Tier
Nuclear control	Catastrophic	IEC 61508 SIL3/SIL4	Tier 1
Military / Defense	Catastrophic	CNSA 2.0, Common Criteria EAL4+	Tier 1
Class III Medical Implants	Critical	FDA Class III, EU MDR Class III	Tier 1
Avionics (DO-178C DAL A)	Catastrophic	DO-178C, DO-254	Tier 1
Class II Medical Devices	Major	FDA PCCP, EU MDR Class IIb	Tier 2
Industrial Control (SIL2)	Significant	IEC 61508 SIL2	Tier 2
Automotive (ASIL B/C)	Moderate/Significant	ISO 26262, UNECE WP.29	Tier 2
Financial Infrastructure	Major	PCI DSS, SOX	Tier 2

Table 7: Tier selection guidance by deployment domain

Part VI: Safety Evidence

17. FMEDA Summary

The Failure Modes, Effects, and Diagnostic Analysis (FMEDA) for the AMAGI TCB demonstrates that the architecture meets the hardware architectural constraints for IEC 61508 SIL2. The analysis covers all components within the TOE boundary: the SCC finite state machine, the MPE enforcement logic, the MAG combinational lookup, the ATL delay circuitry and audit diode, and the Amagi-Q cryptographic engine. The SHMF is analyzed separately as a reliability mechanism with detective security capability only. Detailed FMEDA data is provided in Appendix B.

Parameter	Value	IEC 61508-2 Requirement
Dangerous Undetected Failure Rate (λ_{DU})	18.9 FIT	$PFH < 10^{-6}/h$ (SIL2 high demand)
Safe Failure Fraction (SFF)	93%	Type A: $SFF \geq 90\%$ for $HFT = 0$ (Table 2); includes CRTM + EGW diagnostic coverage
Average Probability of Dangerous Failure on Demand (PFDavg)	6.9×10^{-5}	10^{-3} to 10^{-2} (SIL2 low demand)
Hardware Fault Tolerance (HFT)	0 (Type A, $SFF \geq 90\%$)	SIL2 with $HFT = 0$ when $SFF \geq 90\%$
Diagnostic Coverage (DC)	$\geq 90\%$	$DC \geq 90\%$ required for SIL2 (medium)
Proof Test Interval (TI)	10 years (de-rating from 20-year mission)	Per IEC 61508-6 Annex B

Table 8: FMEDA summary and SIL2 compliance evidence

18. SIL2 Claims and Path to SIL3

The AMAGI Architectural Class achieves SIL2 compliance through a combination of hardware architectural constraints (Type A component, $SFF \geq 90\%$, $HFT = 0$) and formal verification of the TCB. The SIL2 claim is supported by the FMEDA analysis, the SPIN model checking results for the SCC and MPE+ATL, and the TCB complexity bound of 500 lines of HDL. The path to SIL3 requires the following additional measures: (1) increase Hardware Fault Tolerance to $HFT = 1$ through redundant TTP channels, providing a secondary enforcement path that operates independently of the primary SCC/MPE; (2) increase Diagnostic Coverage to $DC \geq 99\%$ through comprehensive online self-testing of all TCB components, including the MAG lookup path and Amagi-Q cryptographic operations; and (3) perform an independent functional safety assessment by an accredited certification body, as required by IEC 61508-1 for SIL3 claims. These measures are designated for Phase 2 development and are not part of the current AMAGI v2.0 baseline.

19. Performance Validation

The AMAGI architecture introduces deterministic, hardware-enforced security overhead that is validated against the timing requirements of safety-critical applications. The performance measurements are based on a prototype implementation using a Xilinx Artix-7 FPGA, which represents a conservative reference target. ASIC implementations are expected to achieve significantly better performance due to the elimination of FPGA routing overhead and the use of dedicated delay circuitry for the TTP programmable delay elements.

Metric	Value	Context
SCC policy verification latency	1 to 10 μ s	Worst-case including cryptographic verification (Xilinx Artix-7)
MAG policy lookup	< 5 ns	Combinational logic in eFuse/OTP; TCAM-class performance
Audit signing (ML-DSA-44)	Decoupled via async FIFO	No timing disruption of 12 to 35 μ s critical path
ATL memory-level delay range	0.5 ns to 1 μ s	Physically enforced; cannot be shortened by software
SCC processor-level delay range	1 ns to 1 s	Physically enforced; covers safety assertion through actuator authorization
Resource utilization (FPGA)	2,500–3,200 LUTs (safety-critical)	Xilinx Artix-7; TCB only (SCC + MPE + MAG query)
Power consumption (FPGA)	< 1.3 W	TCB only; separate power domain from cognitive clusters

Table 9: Performance validation metrics

Part VII: Formal Verification Evidence

20. SPIN Model Checking Results

The AMAGI TCB is verified using the SPIN model checker, which performs exhaustive state-space exploration of Promela models representing the SCC and MPE+ATL finite state machines. The verification covers both safety properties (nothing bad ever happens) and liveness properties (something good eventually happens). The SPIN verification provides strong assurance that the TCB correctly implements the six invariants under all reachable state combinations, complementing the FMEDA analysis which addresses random hardware failures.

Model	States Explored	Safety Properties	Liveness Properties	Result
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SCC FSM (scc_amagi.pml)	2.59×10^7	SCC-S01..SCC-S09 (9)	SCC-S10,S11 trivial; SCC-L01..L04 counterexamples	9 safety PASS; liveness counterexamples (expected)
MPE+ATL (mpe_atl_amagi.pml)	8 states (compact FSM)	MPE-S01..S06, ATL-S01..S05 (11)	MPE-L01..L02, ATL-L01 (3)	All PASS

Table 10: SPIN model checking results

The SCC safety properties verify: (SCC-S01) every trigger results in a policy evaluation before activation; (SCC-S02) a CUTOFF command results in immediate safe state transition; (SCC-S03) the programmable delay cannot be shortened by software; (SCC-S04) the hardware kill switch is isolated from cognitive interference; (SCC-S05) no path exists from trigger to activation that bypasses the SCC; (SCC-S06) the SCC FSM cannot enter an undefined state. The MPE safety properties verify that every memory access is mediated by the MAG policy, that Deny results in immediate rejection, that the Source ID is never transmitted externally, and that External Trigger level requires valid hardware confirmation. The ATL safety properties verify that the delay is physically enforced, that the audit diode is unidirectional, and that all access events generate signed audit records.

21. Formal Verification Strategy and Roadmap

The current verification approach uses SPIN/Promela for exhaustive model checking of the SCC and MPE+ATL finite state machines. This approach provides strong assurance for the implemented FSM logic but is limited to the specific state machine models that are verified: it does not provide a mechanized proof that the TTP predicate, as a mathematical property, holds for all possible implementations. The roadmap for enhanced verification is structured in two phases.

Phase 1 (Current): SPIN model checking of the SCC and MPE+ATL state machines. This covers all reachable states of the implemented logic and verifies that the six invariants hold under all combinations. The TCB complexity bound of 500 lines of HDL ensures that the state space remains tractable for exhaustive exploration. This phase provides sufficient evidence for EAL2 augmented certification and IEC 61508 SIL2 functional safety assessment.

Phase 2 (Planned): Mechanized proof of the TTP predicate at the specification level using Isabelle/HOL. This phase targets two specific theorem classes: (1) **Sufficiency Theorem:** prove that the TTP predicate (Trigger AND Delay AND Activate) implies each of the six invariants (INV-001 through INV-006) individually, under the assumption that the AI model is a hostile insider. That is, for each INV-k, prove the formal statement: $TTP(t) \text{ AND } \text{Assumption_HostileInsider}(t) \text{ IMPLIES } INV\text{-}k(t)$. (2) **Minimality Theorem:** prove that no proper subset of the six invariants suffices to guarantee safety under the hostile insider assumption. That is, for each INV-k, exhibit a counterexample trace in which all invariants except INV-k are satisfied but a safety violation occurs. This is a specification-level proof, not an implementation-level proof: it demonstrates that any system satisfying the TTP predicate automatically satisfies all six invariants, and that dropping any single invariant creates a real attack

vector. Implementation-level verification (proving that a specific 500-line HDL implementation satisfies the TTP predicate) would require HOL-embedded hardware semantics, which is beyond the current scope and would be pursued only if regulatory requirements demand it for higher assurance levels (EAL5+ or SIL3+).

Part VIII: Regulatory and Standards Mapping

22. EU AI Act Annex IV Compliance Mapping

The following table maps each of the nine required sections of EU AI Act Annex IV (technical documentation for high-risk AI systems) to the corresponding sections of the AMAGI Framework v2.0 and its normatively referenced component specifications. Regulation (EU) 2024/1689 requires full compliance by August 2, 2026 for high-risk AI systems. The AMAGI architecture addresses these requirements through its hardware-enforced structural guarantees, which provide a stronger basis for compliance than software-only approaches.

Annex IV Section	Requirement	AMAGI Coverage
1	General description of the AI system	Part I: Sections 1, 4, 5, 7 (invariants, TTP predicate, substrate independence)
2(b)	Design specifications	Part IV: Sections 14, 15 (SFRs, SARs); Part V: Dual-Tier Trust Model
2(c)	System architecture	Part II: Section 8 (Multi-Component TOE); NIPU v1.1; MEM v1.1
2(d)	Data governance	INV-003 (Separation), INV-005 (Mediation): MPE enforces data access boundaries per MAG policy
2(e)	Human oversight	INV-005 (Complete Mediation): hardware-enforced TTP chain; External Trigger (MPE Level 5) provides hardware 2FA for critical operations
2(f)	Predetermined changes	Part V: Dual-Tier Trust Model; Tier 2 PCCP-compliant OTA with monotonic counters
2(h)	Cybersecurity measures	INV-001 through INV-006; PQC suite (FIPS 203/204/205); audit trail (INV-006)
3	Monitoring, functioning, and control	INV-006 (Audit Integrity); FMEDA (Part VI); SPIN verification (Part VII)
5	Risk management system	Part III: Sections 6, 10, 11 (Threat Model, OSPs, Assumptions); Part IV Rationale
9	Post-market monitoring	OE.MONITOR; INV-006 audit trail provides continuous monitoring evidence

Table 11: EU AI Act Annex IV compliance mapping

23. NIST AI RMF Profile Mapping

The NIST AI Risk Management Framework (AI RMF 1.0) organizes risk management activities into four functions: Govern, Map, Measure, and Manage. The following table maps each function to the corresponding AMAGI architectural mechanisms, demonstrating that the framework provides structural (rather than procedural) implementation of the NIST AI RMF requirements.

AI RMF Function	AMAGI Realization	Mechanism
GOVERN	SCC governance of all safety-critical decisions	INV-001: SCC is the sole authority for safety policy; INV-004: MAG immutability ensures governance persistence
MAP	Threat model and risk identification	Part III: AI as Hostile Insider threat model; T.COERCE through T.EXFIL taxonomy
MEASURE	Quantitative safety evidence	FMEDA ($\lambda_{DU} = 18.9$ FIT component / ~ 15.75 FIT system, SFF = 91.6% component / 93% system, $PFD_{avg} = 6.9 \times 10^{-5}$); SPIN model checking (9 safety + 2 assumptions + 4 liveness in 25,927,680 states)
MANAGE	Risk mitigation through architectural controls	Dual-Tier Trust Model; TTP Triad; INV-001 through INV-006 as structural risk controls

Table 12: NIST AI RMF function mapping

24. Post-Quantum Cryptographic Suite and CNSA 2.0 Gap Analysis

The AMAGI Architectural Class mandates the use of post-quantum cryptographic algorithms standardized by NIST in August 2024. All three PQC standards (FIPS 203, FIPS 204, FIPS 205) are final and effective. The following analysis examines the algorithm selection for each TOE function, the security strength justification, and the gap relative to the CNSA 2.0 requirements for National Security Systems.

Function	Algorithm	Standard	Security Level	Signature Size	CNSA 2.0
Key encapsulation	ML-KEM-1024	FIPS 203	256-bit	N/A (KEM)	Compliant
Audit log signing	ML-DSA-44	FIPS 204	128-bit	~2,420 bytes	Gap (requires ML-DSA-87)
NSS digital signatures	ML-DSA-87	FIPS 204	256-bit	~4,627 bytes	Compliant
Auxiliary hashing	SLH-DSA-SHA2-128f	FIPS 205	128-bit	~7,856 bytes	Supplementary

Table 13: PQC algorithm selection with CNSA 2.0 gap analysis

Security Strength Justification for ML-DSA-44 in Audit Signing: The selection of ML-DSA-44 (128-bit security level) for audit log signing is justified by the following analysis. First, the audit log is a detective control (INV-006), not a preventive control: it provides evidence of events that have already occurred, rather than preventing events from occurring. A successful cryptographic attack on the audit log would allow an adversary to forge audit entries, but it would not bypass the TTP enforcement chain (INV-001) or the MAG policy (INV-004). The 128-bit security level provides approximately 2^{128} operations to forge a signature, which is computationally infeasible with both classical and quantum computers for the foreseeable future. Second, the audit FIFO bandwidth is constrained by the memory subsystem architecture: ML-DSA-44 signatures are approximately 2,420 bytes, while ML-DSA-87 signatures are approximately 4,627 bytes, an increase of 91%. For high-frequency audit logging scenarios (e.g., memory access events in a busy SoC), this size difference has a direct impact on FIFO depth requirements and memory allocation. Third, the AMAGI architecture provides a clear upgrade path: deployments requiring CNSA 2.0 compliance can configure Amagi-Q to use ML-DSA-87 at the cost of increased signature size and slightly longer signing latency, while commercial deployments can use ML-DSA-44 for optimal audit throughput.

CNSA 2.0 Migration Path: For National Security Systems (NSS) deployments, the AMAGI architecture supports a dual-parameter configuration: ML-DSA-44 for routine audit log signing and ML-DSA-87 for policy authentication and OTA update signing, where the CNSA 2.0 requirement for 256-bit security is mandatory. This dual-parameter approach provides compliance flexibility: the same hardware (Amagi-Q engine) supports both parameter sets, and the configuration is determined at fabrication time based on the deployment tier. The NSA timeline requires exclusive use of CNSA 2.0 algorithms for firmware and software signing by 2030 and for all NSS categories by 2033. The AMAGI architecture is designed to meet these timelines by supporting ML-DSA-87 as a first-class configuration option from the initial v2.0 release.

25. Harmonised Standards and Cross-References

The following table identifies the harmonised standards and technical specifications referenced by this Framework, their relationship to the AMAGI Architectural Class, and the specific sections that provide compliance evidence.

Standard	Relationship to AMAGI	Framework Section
ISO/IEC 15408 v3.1 (Common Criteria)	PP structure; SFR/SAR selection; EAL2 augmented	Parts III, IV; Sections 14, 15
IEC 61508:2010	SIL2 safety evidence; FMEDA; hardware fault tolerance	Part VI; Sections 17, 18
ISO/IEC/IEEE 15289:2019	Life-cycle documentation structure; information item types	Part II (by reference to NIPU/MEM specs)
FIPS 203 (ML-KEM)	Key encapsulation algorithm conformance	Section 24; MEM v1.1 Section 3.5

FIPS 204 (ML-DSA)	Digital signature algorithm conformance	Section 24; MEM v1.1 Section 3.5
FIPS 205 (SLH-DSA)	Auxiliary hash algorithm conformance	Section 24; MEM v1.1 Section 3.5
FIPS 140-3	Cryptographic module validation alignment (Level 2/3)	Section 15 (FCS requirements)
NIST SP 800-208	Stateful hash-based signatures for firmware signing (bridge technology)	Section 24 (CNSA 2.0 migration)
CNSA 2.0 (NSA)	PQC requirements for National Security Systems	Section 24 (gap analysis and dual-parameter path)
EU AI Act (2024/1689)	Technical documentation for high-risk AI; Annex IV	Section 22; Appendix E

Table 14: Harmonised standards and cross-references

Part IX: S-APL Governance

26. Strict Architectural Principle License (S-APL) v2.0

The Strict Architectural Principle License (S-APL) v2.0 is the legal instrument that protects the structural guarantees of the AMAGI Architectural Class in safety-critical domains. Unlike open-source licenses (MIT, GPL) that govern source code distribution, the S-APL governs the enforced architectural principle: it licenses the right to implement the TTP Triad within any hardware substrate (FPGA, ASIC, SoC security island), subject to compliance with the structural requirements that ensure non-bypassability. The S-APL is not a source code license; it is an architectural compliance license.

The S-APL v2.0 introduces the following changes relative to v1.0: (1) the Derivative Architecture Clause, which requires that any implementation claiming AMAGI-class compliance must maintain all six invariants, and that removal or weakening of any invariant constitutes a derivative architecture that is not covered by the license; (2) the Dual-Tier Compliance Clause, which requires that the deployer select and document the applicable trust tier (Tier 1 or Tier 2) at deployment time, and that any deviation from the selected tier procedures voids the structural guarantee; and (3) the Audit Evidence Clause, which requires the deployer to provide Amagi and relevant regulators with access to the cryptographically signed audit trail as evidence of ongoing compliance.

27. Derivative Architecture Clause

A Derivative Architecture is any implementation that incorporates one or more elements of the AMAGI Architectural Class but does not enforce all six invariants. Derivative Architectures are not covered by the S-APL structural guarantee and may not be marketed, sold, or represented as AMAGI-class systems. The following specific derivative scenarios are addressed:

- **Reduced Assurance Mode:** An implementation that drops one invariant (except INV-001, INV-005, or INV-006) enters Reduced Assurance Mode. Such a system is not AMAGI-class and must be clearly labeled as a reduced-assurance derivative. The S-APL does not prohibit such

implementations but requires that they not be represented as AMAGI-class.

- **Software-Only Implementation:** An implementation that enforces the TTP predicate in software (e.g., within an OS kernel or hypervisor) without hardware enforcement of the delay and activation stages is a derivative architecture. It cannot claim INV-001 (Physical Non-Bypassability) and is therefore not AMAGI-class. The S-APL permits educational and research use of such implementations but prohibits commercial deployment under the AMAGI trademark.
- **Tier Violation:** A Tier 2 system that does not follow the Physical-Presence-Verified OTA protocol, or a system that claims Tier 1 compliance but has an accessible maintenance switch, is in violation of the S-APL Dual-Tier Compliance Clause. Such a system is not AMAGI-class regardless of its other security properties.

28. Compliance Verification Requirements

Licensees under the S-APL v2.0 are required to submit compliance evidence at three stages: (1) Design Compliance, which verifies that the HDL implementation of the TCB satisfies the 500-line complexity bound and that the SPIN model checking results confirm all safety and liveness properties; (2) Deployment Compliance, which verifies that the physical implementation maintains the trust boundaries defined in Section 8.3, that the MAG has been correctly programmed into eFuse/OTP, and that the selected trust tier (Tier 1 or Tier 2) is documented and enforced; and (3) Operational Compliance, which requires periodic audit trail review and, for Tier 2 systems, verification that all policy updates were performed in accordance with the Dual-Tier Trust Model procedures. Compliance verification is performed by AGIAM Technologies Ltd or by an accredited independent assessor designated by AGIAM.

Rationale

R.1 Threat → Objective Traceability Matrix

The following matrix demonstrates that every threat identified in Section 6 is countered by at least one security objective for the TOE, and that every objective contributes to countering at least one threat. This bidirectional completeness is required by ISO/IEC 15408-1 Annex B for Protection Profile conformance.

	O.NONBY PASS	O.MINIMA L_TCB	O.SEPARA TION	O.IMMUT ABLE	O.MEDIA TION	O.AUDIT
T.COER CE	X	X	X			
T.BYPA SS	X				X	
T.TAMP ER	X			X		

T.EXTO RT	X					X
T.ROLL BACK				X		
T.EXFIL			X		X	X

Table 15: Threat to Objective traceability matrix

R.2 Objective → SFR Traceability Matrix

The following matrix demonstrates that every security objective for the TOE is satisfied by at least one SFR, and that every SFR contributes to at least one objective. This bidirectional completeness ensures that the SFR set is both necessary and sufficient.

	FDP	FPT	FIA	FCS	FAU
O.NONBY PASS	IFC.1	PHP.1, SEP.1, FLS.1, RVM.1		CKM.4	
O.MINIM AL_TCB		TST.1			
O.SEPAR ATION	ACC.1, ACF.1, RIP.1	SEP.1			
O.IMMUT ABLE	IFC.1	PHP.1, PHP.3		CKM.1, COP.1	
O.MEDIA TION	ACC.1, ACF.1	RVM.1	UID.1, UAU.1		
O.AUDIT	SDI.1			CKM.1, COP.1	GEN.1, STG.1, STG.3

Table 16: Objective to SFR traceability matrix

R.3 SAR Adequacy Justification

The selection of EAL2 augmented (rather than EAL4+) as the assurance package for the AMAGI Architectural Class is justified by the following considerations. First, the TCB complexity bound of 500 lines of HDL means that the entire security enforcement logic is amenable to exhaustive formal verification through SPIN model checking, which provides stronger assurance than the black-box testing required at EAL3 and EAL4. The SPIN verification covers all reachable states (2.59×10^7 for the SCC) and verifies both safety and liveness properties, which goes beyond the functional testing requirements of EAL4. Second, the augmentation with ADV_SPM.1 (Formal Security Policy Model) ensures that the TTP predicate is formally specified and verified, providing the same level of security policy modeling

assurance as EAL5+ at a fraction of the evaluation cost. Third, the augmentation with ALC_DVS.1 (Development Security) ensures that the design and implementation of the TTP Triad are protected from unauthorized disclosure or modification, which is essential for a hardware security IP core that is delivered as a Hard Macro. Fourth, for deployments requiring higher assurance (e.g., military, nuclear), the AMAGI architecture is designed to be evaluable at EAL4+ or higher in future certification iterations, once the Isabelle/HOL mechanized proof is completed and the testing regime is expanded.

R.4 Architectural Decision Records

The following Architectural Decision Records (ADRs) document the key design decisions that have shaped the AMAGI Framework v2.0. Each ADR captures the context, decision, consequences, and status of a significant architectural choice.

AD R	Decision	Rationale	Status
AD R-0 01	TTP Triad as the foundational security primitive	Hardware-enforced temporal separation provides non-bypassability that software-only approaches cannot achieve; the three-stage mechanism (Trigger, Delay, Activate) maps directly to the six AMAGI invariants	Ratified
AD R-0 02	Dual-Tier Trust Model for policy immutability	Resolves the tension between INV-004 (immutability) and operational need for policy updates (FDA PCCP); Tier 1 for highest assurance, Tier 2 for regulated update capability	Ratified
AD R-0 03	PQC algorithm selection: ML-KEM-1024, ML-DSA-44/87, SL H-DSA-SHA2-128f	FIPS 203/204/205 conformance; ML-DSA-44 for audit throughput (128-bit sufficient for detective control); ML-DSA-87 as CNSA 2.0 option for NSS deployments; dual-parameter Amagi-Q configuration	Ratified
AD R-0 04	Dissolution of ICG and TDF as independent entities	ICG functions absorbed into free inter-cluster communication + MAG routing; TDF functions integrated into SCC FSM as the TTP predicate evaluation engine; reduces TCB surface area and eliminates redundant components	Ratified
AD R-0 05	EAL2 augmented (not EAL4+) for initial certification	TCB complexity bound (500 lines HDL) makes SPIN exhaustive verification stronger than EAL4 black-box testing; ADV_SPM.1 augmentation provides formal policy modeling; future EAL4+ certification path is preserved	Ratified

AD R-0 06	EGW Introduction and EXE Cluster Demotion to Normal World	The EXE cluster is demoted from Secure World to Normal World as a cognitive execution engine; the Executive Gateway (EGW), a non-programmable AND-gate (Actuator_Power = SCC_Enable AND Ext_2FA_Enable), is introduced as the sole hardware actuation gate in the Secure World; eliminates the architectural contradiction of a cognitive cluster residing in the Secure World; strengthens INV-003 by ensuring all cognitive components are outside the TOE boundary	Ratified
AD R-0 07	CRTM (Core Root of Trust for Measurement) Introduction	A mask-ROM-based Core Root of Trust for Measurement (CRTM) is introduced in the Secure World to verify boot integrity of the SCC FSM before the first policy evaluation; the CRTM provides the hardware root of trust for the AMAGI boot chain and post-silicon MAG validation (Golden Hash comparison); approximately 1.4 FIT total failure rate with 95% diagnostic coverage	Ratified
AD R-0 08	Audit Aggregation under Duress	When the Audit FIFO exceeds 50% capacity, Amagi-Q transitions from per-event ML-DSA-44 signatures to Merkle Root batch signatures (one signature per N events, batch interval 10 ms); when FIFO exceeds 90%, a hardware Rate Limiter throttles the MPE bus to prevent CUTOFF in life-critical deployments; preserves INV-006 (audit integrity) while preventing DoS-induced system shutdown per IEC 60601-1 Single Fault Condition	Ratified

Table 17: Architectural Decision Records

Appendices

A. Glossary and Acronyms

Term	Definition
AMAGI	Architectural class for hardware-enforced AI safety systems defined by six invariants and the TTP Triad
ATL	Access Trigger Layer; implements TTP delay at the memory level and the audit transaction logger (hardware diode)
CNSA 2.0	Commercial National Security Algorithm Suite 2.0; NSA PQC mandate for National Security Systems
FMEDA	Failure Modes, Effects, and Diagnostic Analysis; quantitative safety evidence for SIL claims
MAG	Memory Access Graph; 128x128 eFuse/OTP matrix storing 4-bit policy codes governing all data flow permissions
MPE	Memory Policy Enforcer; validates all memory access requests against MAG policy

NIPU	Neuro-Immune Processing Unit; processor-level instantiation of the AMAGI class
PCCP	Predetermined Change Control Plan; FDA framework for medical device software updates
PQC	Post-Quantum Cryptography; algorithms resistant to quantum computing attacks
S-APL	Strict Architectural Principle License; legal instrument governing AMAGI-class implementations
SCC	Security Council Core; minimal TCB (less than or equal to 500 lines HDL) enforcing TTP at the processor level
SFF	Safe Failure Fraction; proportion of safe failures among all failures, used for SIL architectural constraints
SHMF	Self-Healing Memory Fabric; ECC and hash chain-based data integrity with self-isolation capability
SIL	Safety Integrity Level; IEC 61508 measure of safety function reliability (SIL1 to SIL4)
TCB	Trusted Computing Base; the minimal set of hardware and software that must be trusted for security
TOE	Target of Evaluation; the system boundary under Common Criteria evaluation
TTP	Trigger-Programmable Delay-Localized Activation; the foundational security primitive of AMAGI
UCB	Untrusted Computing Base; the AI model and cognitive components treated as potentially hostile

Table A-1: Glossary and acronyms

B. FMEDA Component-Level Breakdown

This appendix provides the component-level FMEDA breakdown for the AMAGI TCB. The aggregate values reported in Section 17 ($\lambda_{DU} = 18.9 \text{ FIT}$, $\text{SFF} = 93\%$, $\text{PFD}_{\text{avg}} = 6.9 \times 10^{-5}$) are the sum of the individual component contributions. This decomposition is essential for SoC integration certification, where an evaluator may require component-level evidence to verify that the failure modes of each subsystem have been individually characterized and that no single component dominates the dangerous undetected failure budget. The component breakdown follows the trust boundary architecture defined in Section 8.3: NIPU-SCC (Trust Boundary 1, source side), MEM subsystem (MPE, ATL, MAG, Trust Boundaries 1 and 2, sink side), and the shared Amagi-Q cryptographic engine (Trust Boundary 3).

Component	Subsystem	λ_D (FIT)	λ_U (FIT)	DC (%)	SFF (%)
SCC FSM core	NIPU v1.1	4.8	0.5	90	92
MPE enforcement logic	MEM v1.1	8.2	2.3	72	88
MAG combinational lookup	MEM v1.1	5.1	1.4	73	89

ATL delay + audit diode	MEM v1.1	3.6	0.8	78	91
Amagi-Q crypto engine	Shared (TB3)	12.4	2.8	77	90
SHMF (ECC + hash chains)	MEM v1.1	6.7	1.1	84	93

Table B-1: FMEDA component-level dangerous failure rates

Parameter	Value	IEC 61508-2 Requirement
Component lambda_DU (sum of parts)	8.9 FIT	Sum of component lambda_DU values
Component lambda_D (sum of parts)	40.8 FIT	Sum of component lambda_D values
System lambda_DU (with CCF)	18.9 FIT	Includes 10.0 FIT common-cause factor per IEC 61508-6
Aggregate SFF	93%	Type A: SFF $\geq$ 90% for HFT = 0 (Table 2); includes CRTM + EGW diagnostic coverage
PFDavg (system)	6.9×10^{-5}	10^{-3} to 10^{-2} (SIL2 low demand)
SIL claim	SIL2 (HFT=0, Type A)	SFF $\geq$ 90% allows SIL2 with HFT = 0
Proof Test Interval	10 years	De-rated from 20-year mission life

Table B-2: FMEDA aggregate and SIL2 compliance evidence

The dominant contributor to lambda_DU is the Amagi-Q cryptographic engine (2.8 FIT, 14.8% of total), driven by the complexity of the ML-KEM-1024 key encapsulation and ML-DSA-44/87 signature verification datapaths. The MPE enforcement logic contributes the second-largest share (2.3 FIT, 12.2%), reflecting the multi-stage access validation pipeline. The ATL audit diode contributes the smallest share (0.8 FIT, 4.2%) due to its simple unidirectional hardware design with minimal combinational logic. The aggregate SFF of 93% exceeds the IEC 61508-2 Table 2 requirement of 90% for SIL2 with HFT = 0 (Type A component), providing a 2% margin against component aging and environmental stress factors. The Proof Test Interval of 10 years is derived from a de-rating analysis of the 20-year mission life, assuming continuous operation at the rated temperature and voltage. For Tier 2 deployments with periodic policy updates, the maintenance events provide additional proof test opportunities that further reduce the effective PFDavg.

B.1 CCF Mitigation: Deep Trench Isolation and Separate Power/Clock Domains

The Common-Cause Failure (CCF) budget of 10.0 FIT (Table B-2, row "System lambda_DU (with CCF)") is predominantly attributable to shared power rail and clock domain coupling between the Secure World and the Normal World in monolithic SoC implementations. IEC 61508-6 Table D.2

identifies shared power supplies and shared clock sources as two of the top three CCF contributors for electronic safety systems. To achieve the target CCF beta factor of $\leq 10\%$ (consistent with IEC 61508-6 Table F.2, Route 2_H), the following physical isolation measures are mandatory for monolithic implementations:

- **Deep Trench Isolation (DTI):** The Amagi-Q cryptographic engine power domain must be physically isolated from the Normal World power domain using Deep Trench Isolation (DTI) structures in the silicon substrate. DTI provides electrical isolation between adjacent circuit blocks by creating high-aspect-ratio trenches filled with dielectric material (SiO₂ or polymer), achieving isolation resistance greater than 1 GOhm and capacitance less than 0.1 pF between domains. This prevents IR-drop coupling and substrate noise injection from the Normal World (where the AI model operates as a Hostile Insider) into the Amagi-Q secure domain, mitigating the Differential Power Analysis (DPA) threat identified in the threat model (T.DPA). DTI is a standard technique in monolithic RF/AMS and secure SoC designs, available at all process nodes from 28 nm and below.
- **Separate PLL (Phase-Locked Loop):** The Secure World clock domain must use a dedicated PLL that is physically and electrically independent from the Normal World clock distribution network. A shared PLL creates a CCF vector whereby a single PLL failure simultaneously affects both the SCC FSM timing and the cognitive cluster timing, potentially creating a race condition that violates INV-001. The separate PLL ensures that a clock failure in either domain does not propagate to the other, maintaining the independent operation guarantee required by IEC 61508-2 redundancy analysis. The Secure World PLL operates at a fixed frequency determined by the OTP configuration, eliminating the attack surface of software-configurable clock dividers.
- **Separate LDO (Low-Dropout Regulator):** Each power domain (Secure World, Normal World, AFC) must have a dedicated LDO voltage regulator fed from the main supply rail. A shared LDO creates a CCF vector whereby a voltage regulator failure simultaneously affects both the SCC and the cognitive clusters. The separate LDO design ensures that a voltage transient in the Normal World (caused, for example, by a compromised AI model executing a power virus attack) does not propagate to the Secure World, where it could cause the SCC FSM to malfunction. The LDO for the Secure World domain must provide voltage regulation within $\pm 2\%$ of nominal, compared to the $\pm 5\%$ tolerance acceptable for the Normal World domain. This tighter regulation is required to maintain deterministic SCC FSM timing.

CCF Beta Factor Derivation: With DTI, separate PLL, and separate LDO implemented, the CCF beta factor is estimated at $\leq 10\%$ per IEC 61508-6 Table F.2 (Route 2_H). This estimate is conservative and assumes: (a) no shared test infrastructure between domains, (b) physical separation of at least 100 μm between Secure World and Normal World active circuits, and (c) independent power-on reset for each domain. The resulting 10.0 FIT CCF contribution yields the system-level λ_{DU} of 18.9 FIT reported in Table B-2. Without these isolation measures, the beta factor would increase to approximately 20–30%, and the resulting λ_{DU} would exceed the SIL2 limit, requiring hardware fault tolerance (HFT ≥ 1) and a fundamentally different certification strategy.

C. Cross-Reference Summary

This appendix provides a consolidated cross-reference summary between the three normative documents of the AMAGI Architectural Class: this Framework v2.0, the NIPU v1.1 Specification, and the MEM v1.1 Specification. The cross-references ensure that any statement made in one document about the architecture can be traced to the corresponding statement in the other two documents. This bidirectional traceability is essential for certification: an evaluator must be able to verify that the component specifications are congruent with the Framework and that no contradiction exists between them.

Topic	Framework v2.0	NIPU v1.1	MEM v1.1
6 Invariants	Section 4	Section 1	Section 1
TTP Predicate	Section 5	Section 2	Section 3.3 (ATL)
SCC FSM	Section 8.1 (logical)	Section 2 (physical, 12 states)	N/A
MPE + ATL	Section 8.2	N/A	Sections 3.1, 3.3
MAG	Section 8.2, 8.3 (TB2)	Section 3 (policy query)	Section 3.2
SHMF	Section 8.2	Section 3 (detective only)	Section 3.4
Dual-Tier Trust	Part V (Sections 16-17)	Section 4	Section 4
Amagi-Q	Section 8.3 (TB3), Section 24	Section 5	Section 3.5
FMEDA	Section 17, Appendix B	Section 6	Section 5 (SPIN)
S-APL v2.0	Part IX (Section 26)	Section 7	Section 8

Table C-1: Cross-reference summary across AMAGI normative documents

D. Change Log (v1.4 to v2.0)

Aspect	v1.4 (Dec 2025)	v2.0 (June 2026)
Document type	Conceptual preprint	Protection Profile + Integration Specification
Component references	NIPU v1.0, MEM v1.0	NIPU v1.1, MEM v1.1
ICG / TDF	Present as independent components	Dissolved; functions in SCC FSM and MAG (ADR-004)
Cluster topology	Monolithic UCB	GEN/COM/ANL/EXE typed clusters

MAG	Not defined	128x128 eFuse/OTP, 4-bit, Dual-Tier update model
SHMF	Not mentioned	ECC + hash chains (Phase 1); AES-256/ML-KEM-1024 (Phase 2)
Amagi-Q	Not mentioned	ML-KEM-1024, ML-DSA-44/87, SLH-DSA-SHA2-128f
Dual-Tier Trust	Not present	ADR-002: Tier 1 Hard Assurance, Tier 2 Controlled Assurance
Formal verification	Planned: Isabelle/HOL	SPIN (completed); Isabelle/HOL (Phase 2, spec-level only)
FMEDA / SIL	Not present	18.9 FIT, SFF 93%, PFDavg 6.9×10^{-5} , SIL2 + path to SIL3
PQC nomenclature	Not specified	FIPS 203/204/205 (final Aug 2024); CNSA 2.0 gap analysis
EU AI Act	General reference	Annex IV compliance mapping (9 sections)
CC structure	Not applicable (preprint)	PP structure; SFR/SAR; Rationale with traceability matrices
Assurance level	Not specified	EAL2 augmented (ADV_SPM.1 + ALC_DVS.1)
License	CC BY-NC 4.0 + S-APL v1.0	S-APL v2.0 (Derivative Architecture Clause, Dual-Tier Compliance, ZKP Audit Protocol, Tier 1 Kill Switch Carve-out, Succession Protocol, Dual-Use Disclaimer)

Table D-1: Change log from v1.4 to v2.0

E. EU AI Act Annex IV — Section-by-Section Compliance Evidence

This appendix provides detailed compliance evidence for each of the nine sections required by EU AI Act Annex IV, supplementing the summary mapping in Section 22. The evidence references specific sections of this Framework and the normatively referenced NIPU v1.1 and MEM v1.1 specifications.

- **Section 1 (General Description):** The AMAGI Architectural Class is defined by six hardware-enforced invariants (Part I, Section 4) and the TTP Triad (Part I, Section 5). The system interacts with hardware through the Hard Macro IP Core delivery format (GDSII, SDC, RTL-Wrapper). The hardware platform is substrate-independent (Part I, Section 7), with a reference target of Xilinx Artix-7 FPGA.
- **Section 2(b) (Design Specifications):** The general logic of the system is defined by the TTP predicate (Part I, Section 5) and the SCC FSM (NIPU v1.1, Section 2). Key design choices include the Dual-Tier Trust Model (Part V), the MAG eFuse/OTP storage (MEM v1.1, Section 3.2), and the PQC algorithm selection (Part VIII, Section 24).

- **Section 2(c) (System Architecture):** The Multi-Component TOE architecture (Part II, Section 8) defines the integration of NIPU v1.1 (processor-level) and MEM v1.1 (memory-level) within a shared SoC. Three trust boundaries govern the interaction between components.
- **Section 2(d) (Data Governance):** Data access is governed by the MAG policy matrix (MEM v1.1, Section 3.2), which enforces typed cluster boundaries (GEN, COM, ANL, EXE). The MPE (MEM v1.1, Section 3.1) mediates all data access requests. The SHMF (MEM v1.1, Section 3.4) ensures data integrity.
- **Section 2(e) (Human Oversight):** The TTP Triad provides hardware-enforced human oversight through the External Trigger enforcement level (MPE Level 5), which requires a hardware 2FA signal for critical operations. The Dual-Tier Trust Model (Part V) requires physical presence for policy updates.
- **Section 2(f) (Predetermined Changes):** The Tier 2 Controlled Assurance model (Part V, Section 16.2) is compatible with FDA PCCP. Policy updates require PQC-signed packages with monotonic counters, verified by Amagi-Q, with hardware fallback to the minimal-safe state.
- **Section 2(h) (Cybersecurity):** All six invariants (Part I, Section 4) provide structural cybersecurity guarantees. The PQC suite (Part VIII, Section 24) provides quantum-resistant cryptographic protection. The audit trail (INV-006) provides forensic evidence of security events.
- **Section 5 (Risk Management):** The threat model (Part I, Section 6) identifies six threat categories. The OSPs (Part III, Section 10) define mandatory security policies. The traceability matrices (Rationale) demonstrate complete coverage of all threats by objectives and SFRs.
- **Section 9 (Post-Market Monitoring):** The INV-006 audit trail provides continuous, tamper-proof monitoring evidence. The OE.MONITOR objective (Part IV, Section 13) requires post-market monitoring per EU AI Act Article 72. The FMEDA (Part VI, Section 17) provides quantitative reliability evidence.

H. ADR Register

The Architectural Decision Record (ADR) register provides a comprehensive list of all significant architectural decisions made during the development of the AMAGI Framework. The current register contains eight ratified ADRs (ADR-001 through ADR-008). ADR-001 establishes the TTP Triad as the foundational security primitive. ADR-002 introduces the Dual-Tier Trust Model. ADR-003 defines the PQC algorithm selection with security strength justification (128-bit for audit, 256-bit for NSS). ADR-004 documents the dissolution of ICG and TDF. ADR-005 ratifies the EAL2 augmented assurance level for initial certification. ADR-006 introduces the Executive Gateway (EGW) and demotes the EXE cluster to Normal World, resolving the TOE boundary contradiction. ADR-007 introduces the CRTM (Core Root of Trust for Measurement) for boot integrity verification and post-silicon MAG validation. ADR-008 introduces Audit Aggregation under Duress (Merkle batch signing and Rate Limiter) to prevent DoS-induced CUTOFF in life-critical deployments. Future ADRs will address Phase 2 extensions (hardware-accelerated encryption, SIL3 path), the Isabelle/HOL verification strategy (Sufficiency Theorem and Minimality Theorem per Section 21), and the ASIC reference

implementation. All ADRs are summarized in Table 17 of the Rationale section.

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